

Ref. No: CUIET/CSE/ACAD/2026/95

Date: 02.04.2026

NOTICE

Test Type: ST

Test No: 1

Total Marks: 40

Duration: 90 minutes

Name of Course Coordinator: Dr. Shivani Wadhwa

Course Code: 22CS036

Course Title: Advanced Data Structures

Day: Tuesday

Date: 07-04-2026

Time: 9:00 am to 4:00 pm

Mode: Online (On Campus)

Program & Semester: B.E.(CSE) & 6th Sem

Platform: Testpad

Group: G1 to G14, G15, G16, G17 to G27, G31, G32, G34, G35, G36

Syllabus for Test: Arrays – 1: Advance Problems on Sliding Windows and Two pointers, Arrays – 2: Frequency Arrays and Prefix Arrays and Its Problems, Binary Search: Binary Search, Its Implementation and Advance Binary Search problems, Array Advanced Algorithms – 1: Kadane's Algorithm, Prime Sieve, Vectors and its Amalgamation, Strings: Implementation Problems on Strings, Two pointers in Strings and Problem on Frequency Arrays with respect to Strings, Sliding Window – 1: Advance Problems on Sliding Window and Deque STL, Advanced String Algorithms: String Matching Algorithms like KMP, Z Function, and Rabin Karp, Recursion – 1: Basics of Recursion, basic problems on recursion to have better understanding of the call stack, Types of Recursion, Recursion Backtracking – 1: Implementation based Problems on recursion with call stacks, Introduction to Backtracking and basic problem on backtracking, Recursion Backtracking – 2: Advance problems on backtracking, Recursion on Matrix, OOPS-1: OOPs with real world example why it is important, classes, objects and self invoked functions like Constructors, Copy Constructor, Copy Assignment Operators, Destructors, OOPS-2: Characteristics of OOPs like Encapsulation, Abstraction, Polymorphism, Inheritance, Dynamic Binding, Message Passing etc, Linked List.

Format of Assessment:

10 MCQs: 1 Mark

5 MCQs: 2 Marks

2 Coding: 5 Marks

1 Coding: 10 Marks

Total= 40 Marks

Signature of Course Coordinator:

